

Interpreting a Neural Network Model of Exoplanet Atmospheric Abundances: A Multi-Method Feature Attribution Analysis

Abstract

Neural networks can retrieve chemical abundances from exoplanet transmission spectra with high accuracy, but their internal reasoning is hidden, which limits the reliability of results. We train a multi-layer perceptron (MLP) to predict the abundances of water (H₂O) and methane (CH₄) from simulated atmospheric spectra, achieving $R^2 = 0.908$ for H₂O and $R^2 = 0.937$ for CH₄ on a held-out test set. To assess whether the model's predictions are grounded in physically meaningful spectral features, we apply four complementary feature-attribution techniques (SHAP, LIME, and occlusion and permutation importance) and compare their outputs. Directional occlusion and permutation importance agree closely (Spearman $\rho = 0.950$ for H₂O, $\rho = 0.873$ for CH₄), and a four-method consensus analysis identifies a compact set of wavelength bands that consistently drive each species' predictions. These results indicate that the network has learned representations that are both accurate and, to a substantial degree, physically interpretable.

1. Introduction

Neural networks are increasingly used to retrieve atmospheric properties from exoplanet spectra because they learn complex, nonlinear mappings between spectral features and chemical abundances far faster than traditional Bayesian retrieval codes. This speed comes at a cost: a model that achieves high accuracy gives no direct evidence that it has learned the underlying physics rather than a dataset-specific shortcut.

This work trains an MLP to predict H₂O and CH₄ abundances from simulated transmission spectra, then applies four independent feature-attribution techniques (SHAP, LIME, occlusion, and permutation importance) each grounded in a different mathematical basis. Where these methods agree, that agreement is strong evidence that the identified wavelength regions reflect genuine model behaviour rather than an artifact of any single technique. A consensus ranking across all four methods identifies the spectral regions most consistently associated with each species' predictions.

2. Data and Model

The dataset consists of simulated exoplanet transmission spectra for, spanning a true wavelength grid of 0.483–16.917 μm across 100 points. Each sample includes a forward-model spectrum and abundances for ten chemical species; this study targets H₂O and CH₄. Input spectra were normalised per sample using min-max scaling; target abundances were z-score standardized using training-set statistics only, and the data were split 80/20 into training and test sets.

The model is a feedforward MLP with three hidden layers (512, 256, 128 units), LeakyReLU activations, trained with Huber loss and the Adam optimizer ($\text{lr} = 1 \times 10^{-3}$, batch size 64) for 50 epochs. Training loss fell steadily from 0.0982 to 0.0329 with no early stopping required.

3. Predictive Performance

The model achieved $R^2 = 0.908$ for H₂O and $R^2 = 0.937$ for CH₄ on the held-out test set (Figure 1), indicating that it captures the great majority of variance in both target abundances across the simulated planet population.

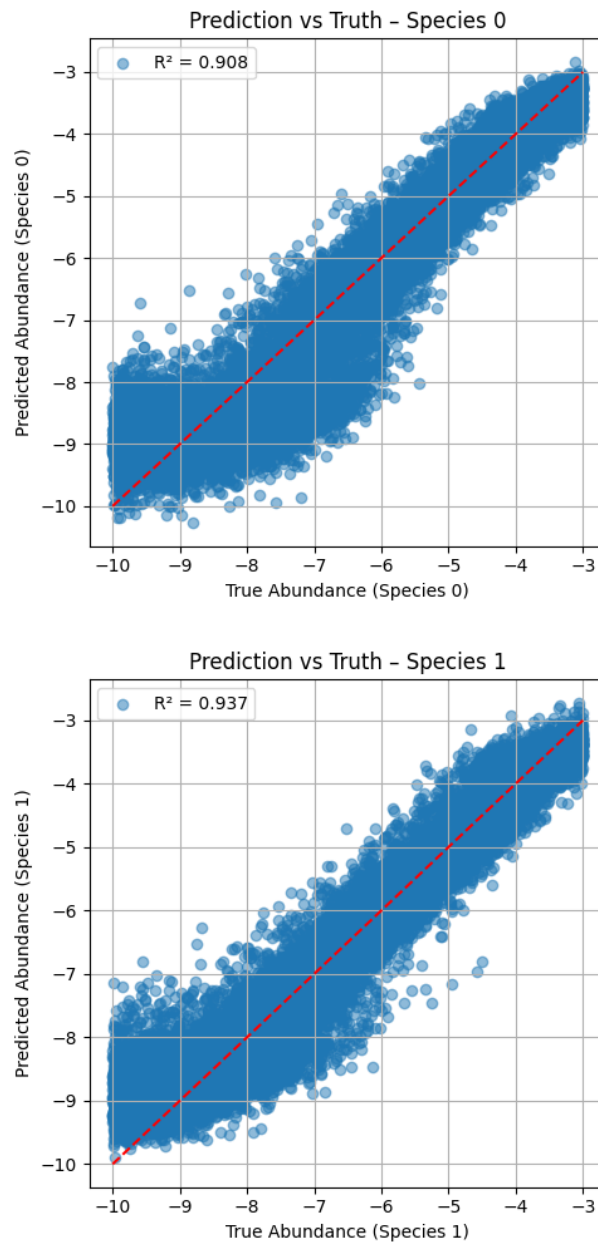


Figure 1. Predicted versus true log-abundance for H₂O (top, $R^2 = 0.908$) and CH₄ (bottom, $R^2 = 0.937$).

4. Feature Attribution Methods

Four attribution methods were applied to identify which wavelengths most influence the model's predictions. SHAP (GradientExplainer) computes game-theoretic contribution scores relative to a training-set background. LIME fits a local linear surrogate model around sampled test spectra. Occlusion importance measures the change in prediction when a local wavelength band is replaced with its training-set mean. Permutation importance measures the change in prediction when a band's values are randomly shuffled across test samples. Both perturbation methods were computed with directional (signed) and magnitude (absolute) variants.

5. Cross-Method Agreement and Consensus

The Spearman rank correlation between directional occlusion and permutation importance was $\rho = 0.950$ ($p = 2.1 \times 10^{-51}$) for H₂O and $\rho = 0.873$ ($p = 2.2 \times 10^{-32}$) for CH₄ (Figure 2), indicating strong, statistically significant agreement between these two independently-derived rankings.

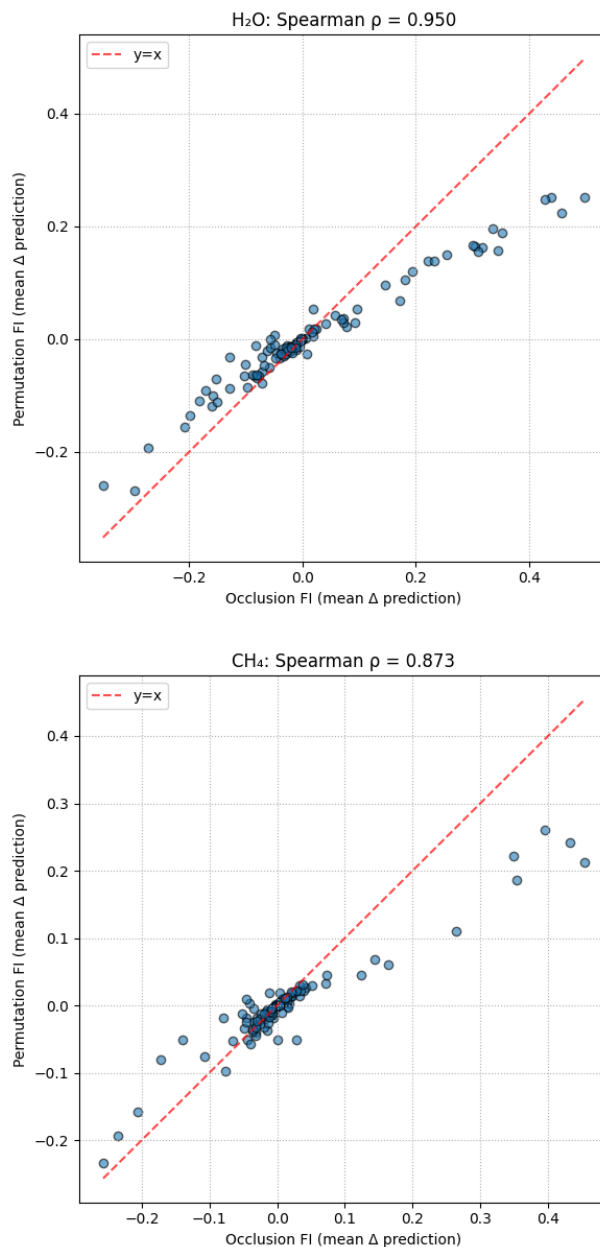


Figure 2. Spearman rank correlation between directional occlusion and permutation importance for H₂O (left, $\rho = 0.950$) and CH₄ (right, $\rho = 0.873$).

A consensus importance score was constructed by ranking wavelength bands within each of the four methods and averaging ranks across methods. The resulting top-5 bands for each species are shown in Table 1 and Figure 3.

Species	Rank	Wavelength Band (μm)
H ₂ O	1	7.48 – 8.48
H ₂ O	2	5.48 – 6.48

H2O	3	2.48 – 3.48
H2O	4	0.48 – 1.48
H2O	5	10.48 – 11.48
CH4	1	3.48 – 4.48
CH4	2	4.48 – 5.48
CH4	3	5.48 – 6.48
CH4	4	0.48 – 1.48
CH4	5	1.48 – 2.48

Table 1. Top-5 consensus wavelength bands for H2O and CH4, ranked by mean importance rank across SHAP, LIME, occlusion, and permutation methods.

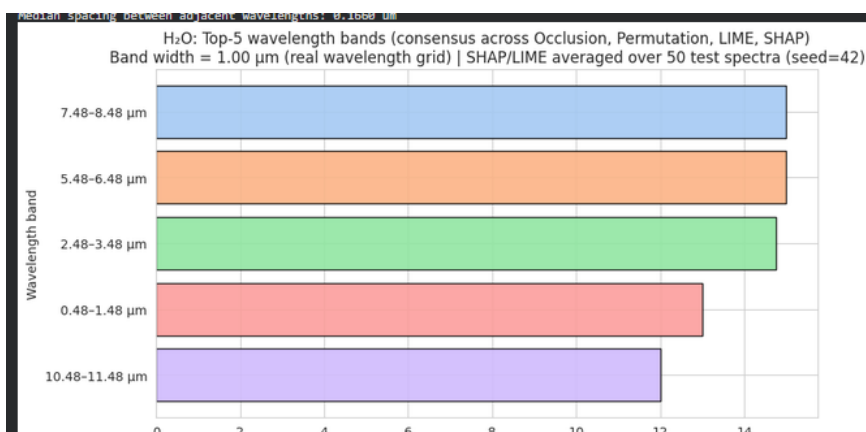


Figure 3a. Top-5 consensus wavelength bands for H2O.

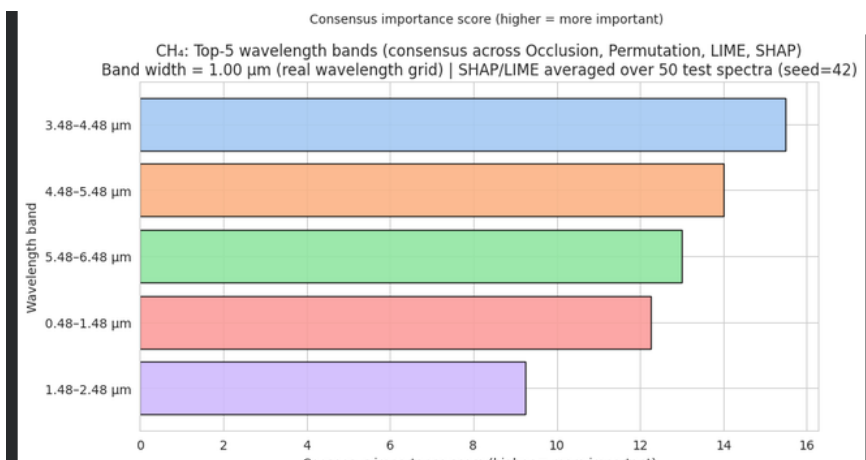


Figure 3b. Top-5 consensus wavelength bands for CH4.

The leading CH4 band (3.48–4.48 μm) lies just past the well-known C–H stretch fundamental near 3.3 μm, lending physical plausibility to the model's learned representation. The H2O bands at 0.48–1.48 and 2.48–3.48 μm are broadly consistent with known near-infrared water absorption structure. The remaining bands, in the 4.5–8.5 μm range for both species, are more difficult to assign to a single diagnostic feature at this bin resolution and may reflect broader, blended opacity structure in the underlying forward model.

6. Discussion

The model's high predictive accuracy is accompanied by a reasoning process that is, to a meaningful extent, recoverable and physically grounded. The strong agreement between occlusion and

permutation importance (two methods with different perturbation mechanisms) argues against the identified wavelength bands being an artifact of any one technique, and the convergence of all four methods on a compact consensus set, including a band close to a known CH₄ feature, supports the interpretation that the network has learned genuine spectral structure rather than a purely statistical shortcut.

The CH₄ correlation between occlusion and permutation ($\rho = 0.873$) is measurably lower than that for H₂O ($\rho = 0.950$), suggesting the model's CH₄ predictions may draw on a somewhat less consistent set of features, or that the CH₄ spectral signature is more entangled with other species in this dataset. A limitation of this analysis is that it was conducted on simulated spectra from a specific forward model; the consensus bands identified here should be validated against independently simulated or real spectra before being treated as general prescriptions for observational campaigns.

7. Conclusion

An MLP trained to predict H₂O and CH₄ abundances from simulated exoplanet transmission spectra achieves strong predictive accuracy ($R^2 = 0.908$ and 0.937 , respectively) while also supporting a consistent, multi-method interpretability analysis. SHAP, LIME, occlusion, and permutation importance converge on a compact set of wavelength bands for each species, with occlusion and permutation importance in particularly strong agreement. This combination of accuracy and interpretability is a necessary step toward the responsible use of neural network retrieval methods in exoplanet atmospheric characterization.